

St David's Marist Inanda

Tour de Maths 2014 Junior Paper

CASIO®

1.) Two gears, one with 12 teeth and the other with 20 teeth, fit together as shown in the illustration. After how many rotations of the smaller gear will the two gears be back in exactly the same position?

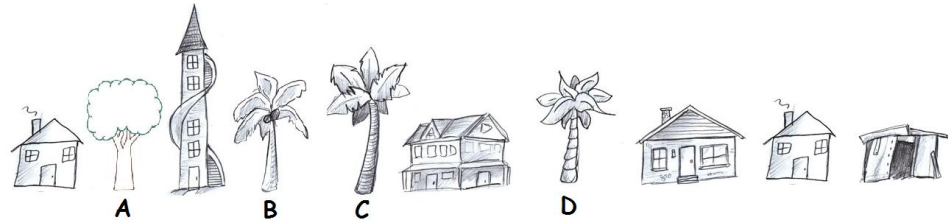


2.) An amoeba propagates by simple division; each split takes 3 minutes to complete. When such an amoeba is put in a glass container with a nutrient fluid, the container is full with amoeba's in an hour. How long would it take for the container to be filled if we started with not 1 but 2 amoeba's?

3.) $\log_{16} 256 = ?$

4.) $\frac{1 \times 9 \times 9 \times 8}{1 + 9 + 9 + 8} = ?$

5.) The trees and the houses in the diagram below are all in a straight line. In each of the six houses lives a child. At which tree should the children meet so that the sum of the distances they walk to that tree is a minimum?

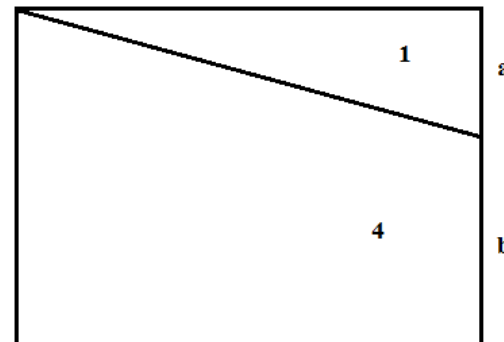


6.) $2^x + 3^y = 41$ If both x and y are natural numbers.
 $x + y = ?$

7.) What was the date 65 days ago?

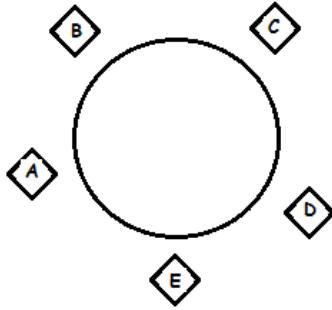
8.) How much time is there between 11/11/11 11:11am and 12/12/12 12:12pm? Give your answer in years, months, days, minutes and seconds.

9.) In the diagram below, the sloping line divides the area of the rectangle in the ratio 1:4 what is the ratio of $a:b$?



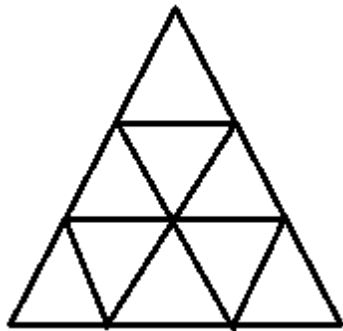
10.) In a contest to guess the number of balloons in a bunch, Sarah guessed 25, Betty guessed 31, Nomsa guessed 29, Jakkie guessed 23 and Edith guess 27. Two guesses were wrong by 2, and two guesses were wrong by 4. The other guess was correct. How many balloons were in the bunch?

11.) In how many different ways can 5 people be seated round a table?



12.) If the date on Thursday is the 29th, what was the date on the 2nd Saturday of the month?

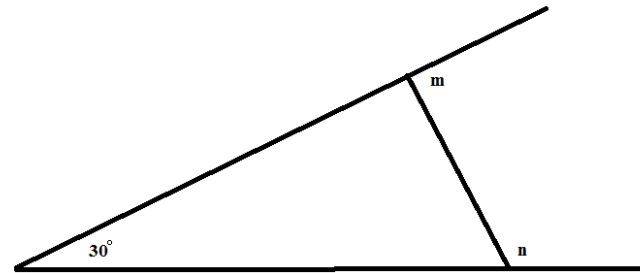
13.) How many triangles are there in the diagram below?



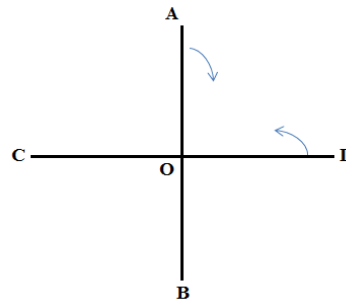
14.) Maggie rolls an ordinary six sided die repeatedly, keeping track of each number as she rolls, and stopping as soon as any number is rolled for the 5th time. If Maggie stops after her 9th roll, and the sum of the numbers she rolled is 20, then how many combinations of numbers could she have rolled?

15.) The desks in a classroom are arranged in straight rows. John is in the third row from the front and the fourth row from the back. He also third from the left end of a row and fifth from the right. How many desks are there in the class?

16.) In the diagram below, the angle m is $\frac{2}{3}$ the size of n . What is the value of angle m ?



17.) Two lines AB and CD intersect at 90° at O and start rotating around O in different directions as shown below. AB rotates at one revolution every two minutes and CD at three revolutions a minute. After how many seconds will the lines be concurrent (on top of each other) for the first time?



18.) What is the maximum number of intersections between any 4 circles?

19.) If every circle below has a radius of 5, what is the height of the entire arrangement?

